

HTML Tables

Lab 3 — Reformatted Notes with Extra Detail on Structure, Attributes & Accessibility

Section	Covers
1	<code><table></code> — the container element and its attributes
2	<code><caption></code> — giving the table a title
3	<code><tr></code> — table rows
4	<code><th></code> vs <code><td></code> — header cells vs data cells
5	<code><thead></code> , <code><tbody></code> , <code><tfoot></code> — grouping rows
6	<code>colspan</code> and <code>rowspan</code> in depth
7	<code>cellpadding</code> and <code>cellspacing</code> — legacy attributes vs CSS
8	Complete worked example — Employee Attendance table
9	Accessibility and nested-table notes (new)
10	Where this fits in the course workflow (HTML -> CSS -> Bootstrap -> Tailwind)
11	Revision quiz

1. <table> — The Container Element

Definition: Defines a table. A table is built from rows, which are in turn built from cells — it's a container, not a cell itself.

Attribute	Meaning
border	Legacy way to draw a visible border around the table and its cells
width	Legacy way to set the table's width; CSS width is preferred today
cellpadding	Space INSIDE each cell, between its content and its border
cellspacing	Space BETWEEN adjacent cells
align (obsolete)	Old way to align the whole table on the page; replaced by CSS margin/float

```
<table border='1'>
  ...
</table>
```

Recap / Extra: All of these attributes are considered legacy/presentational HTML. Modern practice is to use `border='1'` only for quick demos, and to move width, padding, spacing, and alignment into CSS (covered in Section 11).

2. <caption> — Giving the Table a Title

Definition: Gives a visible title/caption to the table. It must be the first child inside <table>, before any row or row-group.

Along with the global attributes every HTML element accepts (id, class, style, title), <caption> takes no attributes of its own.

```
<table>
  <caption>Employee Attendance</caption>
  ...
</table>
```

Example: A <caption> is announced by screen readers before the table content, so it also serves an accessibility role — not just a visual heading.

3. <tr> — Table Rows

Definition: Creates one horizontal row in the table. A row normally contains <th> and/or <td> cells, never text directly.

```
<tr>
  <td>Zeel</td>
  <td>Present</td>
</tr>
```

Recap / Extra: The number of cells in each <tr> should match across rows in the same table section, or the columns will visually misalign.

4. <th> vs <td> — Header Cells vs Data Cells

Tag	Purpose	Default styling
<th>	A heading cell — labels what a row/column represents	Bold and centered
<td>	A normal data cell — holds the actual value	Regular weight, left-aligned

```
<th>Name</th>  
<td>Mann</td>
```

4.1 colspan and rowspan on both tags

Both <th> and <td> accept colspan and rowspan — covered in full in Section 7.

4.2 scope — connecting a header to its cells (new)

scope: An accessibility attribute placed on <th> that tells screen readers whether the header applies to its column (scope='col') or its row (scope='row').

```
<th scope='col'>Name</th>  
<th scope='row'>Monday</th>
```

Recap / Extra: The original notes didn't include scope — it isn't required for a table to render correctly, but it's what makes a data table actually usable with a screen reader, and is a common addition expected in accessible-design questions.

5. <thead>, <tbody>, <tfoot> — Grouping Rows

Tag	Purpose
<thead>	Groups the header row(s) — typically one <tr> of <th> cells
<tbody>	Groups the main data rows — a table can have multiple <tbody> sections
<tfoot>	Groups footer row(s) — totals, summaries, or closing notes

```
<thead>...</thead>
<tbody>...</tbody>
<tfoot>...</tfoot>
```

5.1 Rendering-order rule (new)

Example: Per the HTML5 spec, <tfoot> may be written either before or after <tbody> in the source code, but browsers always render it as the last section of the table (footer at the bottom), regardless of where it appears in the markup.

6. colspan and rowspan in Depth

Attribute	Meaning
colspan	Makes a cell stretch across multiple columns
rowspan	Makes a cell stretch across multiple rows

```
<tr><td colspan='2'>End of Week</td></tr>
```

```
<tr>  
  <td rowspan='2'>Monday</td>  
  <td>Present</td>  
</tr>  
<tr><td>Late</td></tr>
```

Recap / Extra: Memory trick: COLSPAN = columns (spans sideways). ROWSPAN = rows (spans downward). When a cell uses rowspan, the rows below it must have one fewer <td> in that position, or the table will visually shift.

7. cellpadding and cellspacing — Legacy vs CSS

Attribute (legacy HTML)	What it does	CSS equivalent
cellpadding	Space inside each cell, between content and border	padding on td/th
cellspacing	Space between adjacent cells	border-spacing on table (with border-collapse: separate)

```
<table cellpadding='5' cellspacing='2'>
  ...
</table>
```

Example: In CSS, border-collapse: collapse merges adjacent cell borders into one line and disables cellspacing entirely — a detail that trips students up once they move from HTML attributes to CSS.

8. Complete Worked Example — Employee Attendance

Same structure as your original complete example, rebuilt around an Employee Attendance table so it isn't an identical copy.

```
<table border='1' cellpadding='5'>
  <caption>Employee Attendance</caption>
  <thead>
    <tr>
      <th scope='col'>Employee</th>
      <th scope='col'>Department</th>
      <th scope='col'>Status</th>
    </tr>
  </thead>
  <tbody>
    <tr>
      <td>Zeel</td>
      <td>Web Development</td>
      <td>Present</td>
    </tr>
    <tr>
      <td>Yug</td>
      <td>QA</td>
      <td>On Leave</td>
    </tr>
  </tbody>
  <tfoot>
    <tr>
      <td colspan='3'>2 employees shown</td>
    </tr>
  </tfoot>
</table>
```

9. Accessibility and Nested-Table Notes

9.1 Accessibility recap

Three things make a table screen-reader friendly: a `<caption>` describing what the table is, `<th>` (not `<td>`) for every header cell, and `scope` on each `<th>` to say whether it labels a column or a row.

9.2 Avoid nested tables (new)

Example: It's technically possible to put a full `<table>` inside a `<td>`, but this was historically used for page layout (pre-CSS era) and is now considered bad practice — it breaks screen readers and responsive resizing. Use `<table>` only for genuinely tabular data, and CSS (flexbox/grid) for page layout.

10. Where This Fits in the Course Workflow

Following the course progression HTML -> CSS -> Bootstrap -> Tailwind CSS, most of the presentation-related attributes in this lab get replaced as you move forward.

This lab (legacy HTML)	CSS (next lab)	Bootstrap / Tailwind (later labs)
border	border property + border-collapse	Bootstrap: table-bordered Tailwind: border
cellpadding	padding on td/th	Bootstrap: table classes Tailwind: p-2
cellspacing	border-spacing	Tailwind: border-spacing-2
width (on table)	width / max-width	Tailwind: w-full
align (obsolete)	margin: auto / text-align	Tailwind: mx-auto

Recap / Extra: Keep this Lab 3 table exactly as it is for now — the CSS lab will style this same markup, and the Bootstrap/Tailwind labs will restyle it again using framework classes instead of custom CSS.

11. Revision Quiz

1. What must be the first child inside `<table>` if a caption is used?
 2. What is the difference between `<th>` and `<td>` in both purpose and default styling?
 3. What does the `scope` attribute do on a `<th>`, and what are its two common values?
 4. What is the rendering-order rule for `<tfoot>`, regardless of where it's written in the HTML?
 5. What is the difference between `<colgroup>/<col>` and styling each `<td>` individually?
 6. Explain `colspan` vs `rowspan` with one example each.
 7. What is the difference between `cellpadding` and `cellspacing`, and what CSS properties replace them?
 8. Why is putting a full `<table>` inside a `<td>` (a nested table) considered bad practice today?
 9. Which HTML table attributes are considered obsolete/legacy, and what replaces them in CSS?
 10. In the Employee Attendance example, which tag groups the summary row at the bottom?
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